

3 A student investigated the water potential of carrots.

Six cylinders were cut from a single carrot. One cylinder was placed in distilled water and the others were placed in solutions of different concentrations of sucrose.

The length of each cylinder was measured using a ruler every 5 minutes for the next 50 minutes.

(a) (i) Suggest how the method can be modified to increase validity.

(1)

(ii) Suggest how the method could be modified to reduce measurement errors.

(1)

(b) (i) Explain how a 0.4 mol dm^{-3} solution could be made from a 1.0 mol dm^{-3} sucrose solution.

(2)

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(ii) Suggest **two** properties of sucrose that make it suitable for use in this investigation.

(2)

1

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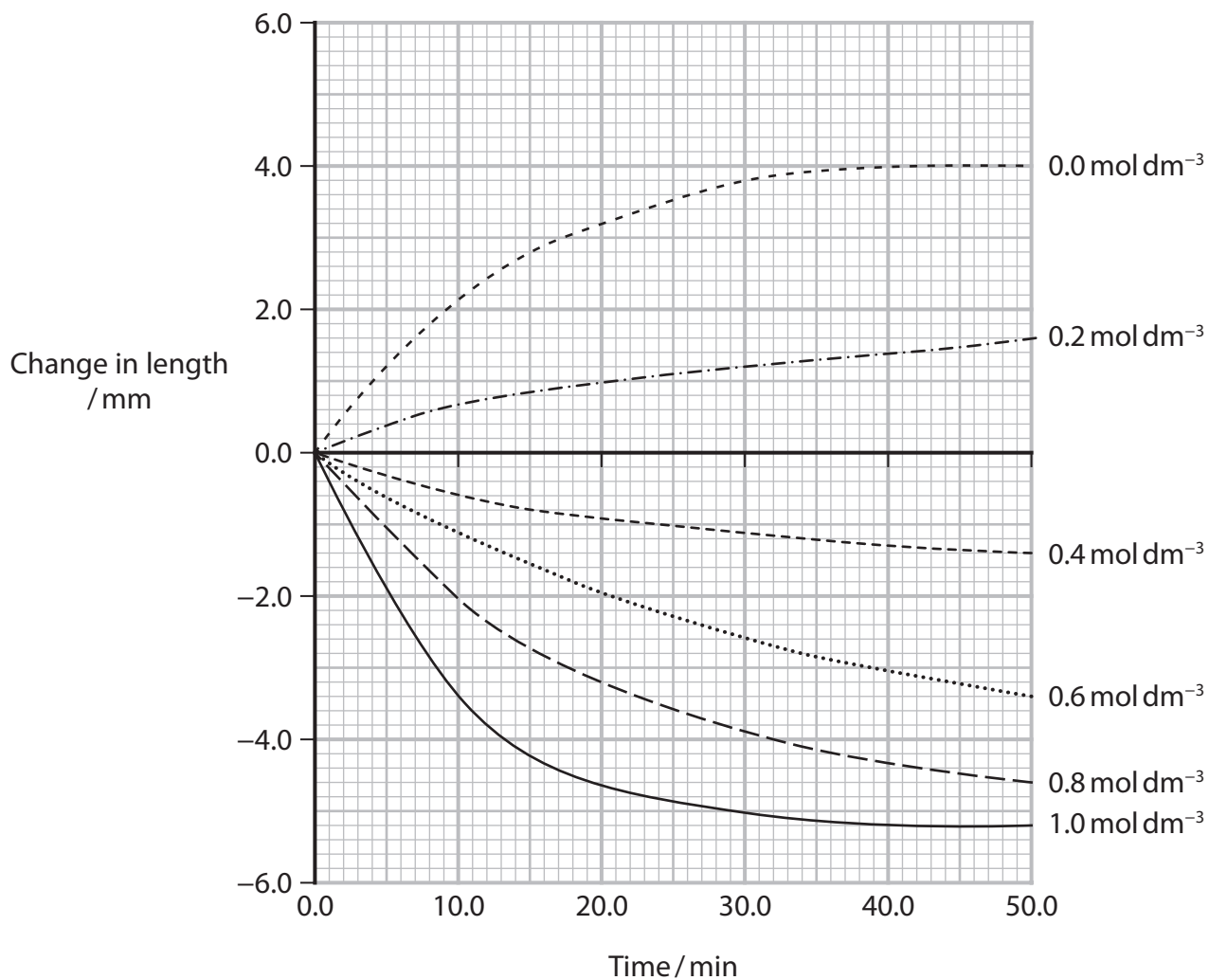
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(c) The graph shows the results of this investigation.



(i) Draw a table to show the results at 50 minutes for all six concentrations.

(3)

- (ii) Calculate the rate of change in length of the carrot cylinder in a 1.0 mol dm^{-3} solution at 15 minutes.

(2)

Answer mm min^{-1}

- (iii) Explain the results between 40 and 50 minutes for the 1.0 mol dm^{-3} solution.

(2)

- (iv) Explain the results between 40 and 50 minutes for the 0.0 mol dm^{-3} solution.

(2)

(Total for Question 3 = 15 marks)

TOTAL FOR PAPER = 50 MARKS

